# Assignment 3 (Possible points: 100)
# Topics: Descriptive Analytics

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**Instructions:**

Refer back to all of the slides and explanations within Weeks 5 and 6 to do this assignment.

**Submit this assignment on BrightSpace. If you have sent the emails, please submit it again on BrightSpace.**

**Due date: Midnight Sunday, March 8, 2026.**
**Only submit Word document (.doc or .docx) or .pdf file.**
**Save the file with your name at the beginning, such as**
**John Doe _ Assignment 3 _ MIS 430.01 / MIS 580N.01 – Spring 2026.docx**

Use the dataset attached, which covers real or fake news on Facebook spanning across multiple years.

All tasks must clearly state which tool(s) were used (such as Excel, Tableau, R ggplot2, R Shiny, etc.) and interpret the key results (such as biggest and smallest records, upward or downward trends or patterns across the timeline). Label your choices clearly, such as Choice #1.

Exception:
Citations and references are not required, but highly encouraged in interpretation and discussions of the results, especially when trying to explain the findings.

Major concerns are the analyses and interpretation.

**Part 1: Basic descriptive statistics**

For both MIS 430 and MIS 580N:

Work on all 3 meaningful numerical variables / columns: share counts, reaction counts and comment counts.

Based on the chosen variables / columns (one at a time) and your understanding, perform **basic descriptive statistics** for the given data, which may include but are not limited to some (not all) of these:

- Compute mean, median, range (min – max), standard deviation, etc.
- Detect missing values and suggest solutions: should we remove records (regarding the size of the samples), or replace with mean, median or other values?
- Detect outliers, and argue if we should ignore or focus on those outliers. You can use either statistical results or visualization (as in the second part) to detect outliner. Again, this is optional.

**Part 1: Data Visualization**

MIS 430 Instruction: Generate **two graphs** chosen from the ideas below. Interpret the meaning of key findings, and discuss the values of such findings. Clearly show at least **most of the labels** of records.

MIS 580N Instruction: Generate **three graphs** chosen from the ideas below. Interpret the meaning of key findings, and discuss the values of such findings. Clearly show at least **most of the labels** of records.

1. A pie chart of the percentages of (1) different types of ratings of falsehood; or (2) different categories; or (3) pages; or (4) post types.
2. A bar chart of the numbers of records belong to (1) different types of ratings of falsehood, or (2) different categories; or (3) pages; or (4) post types.
3. A histogram displaying the distribution of records of one of these: (1) share counts, (2) reaction counts, (3) comment counts.
4. A line chart showing the total numbers of records across published dates.
5. A line chart showing the total numbers of records across published dates, with different lines for one of these choices: (1) different types of ratings of falsehood; or (2) different categories; or (3) pages; or (4) post types.

## <u>Beginning your work from here, retaining previous instructions:</u>

**Part 1: Descriptive statistics: [Tableau]**

Shares:

| | |
|---|---|
| Avg. Share Count | 4,045 |
| Count of Share Count | 2,212 |
| Max. Share Count | 1,088,995 |
| Median Share Count | 96 |
| Min. Share Count | 1 |
| Std. dev. of Share Count | 29,832 |

Reactions:

| | |
|---|---|
| Avg. Reaction Count | 5,364 |
| Count of Reaction Count | 2,280 |
| Max. Reaction Count | 456,458 |
| Median Reaction Count | 546 |
| Min. Reaction Count | 2 |
| Std. dev. of Reaction Count | 19,127 |

Comments:

| | |
|---|---|
| Avg. Comment Count | 516 |
| Count of Comment Count | 2,280 |
| Max. Comment Count | 159,047 |
| Median Comment Count | 132 |
| Min. Comment Count | 0 |
| Std. dev. of Comment Cou.. | 3,569 |

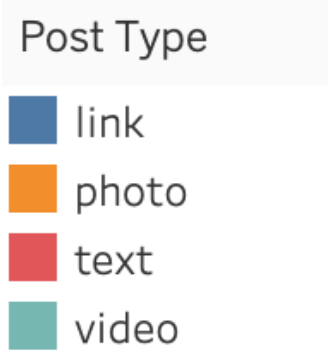
**Descriptive Analysis - Shares:**
This dataset contains 2,212 records for share count. The average number of shares is 4,045, while the median is only 96 shares; this large gap in difference indicates that the distribution is rightly skewed. This means that most posts receive relatively low shares, but there is a small number of posts in the data set that go viral; this is due to the significant weight of a viral post that raises the average. The maximum share count is 1,088,995, which is extremely high compared to the median (96); these large numbers caused by viral posts demonstrate the strength of outliers. Also, the standard deviation is 29,832; this means that the number of shares vary greatly, compared to the average. The large values (max share count, and average) within the data set, when compared to the size of the median demonstrated how meaningful it is to analyze the strength of viral posts; therefore, these outliers should be included with dataset. There are 2,280 total records in the dataset for reactions and comments, however, there are only 2,212 records for share count. This represents 68 missing values, however the missing data only accounts for roughly 3% of the total dataset. Given the large sample size of 2,282 records, removing these 68 incomplete rows would not meaningfully reduce the dataset.

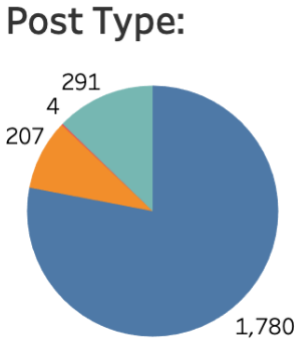
**Descriptive Analysis – Reactions:**
This dataset contains 2,280 records for the reaction count. The average number of reactions is 5,364, while the median is only 546; similar to share count, this large gap indicates a heavily right-skewed distribution. Most posts received modest engagement, while a small number of posts attracted extremely high reaction counts, pulling the average well above the median. The maximum reaction count is 456,458, which is roughly 838 times larger than the median, this illustrates the extreme nature of the outliers. The minimum reaction count is just 2, and the standard deviation confirms that there is enormous variability in how many reactions posts receive. Similar to share count, the large values driven by viral posts are legitimate data points and should be retained in the analysis rather than removed, as they reveal meaningful patterns about which types of content generate the highest levels of audience engagement on Facebook. There are only 2 records that have missing reaction count values (0.09%), which can be ignored due to its low impact on the data.

**Descriptive Analytics – Comments:**
This dataset contains 2,280 records for comment count. The average comment count is 516, while the median is 132. This gap is smaller than those seen in shares and reactions, however this is still an indication of right-skewed distribution where most posts receive relatively few comments and a handful of posts generate extremely high comment activity. The maximum comment count is 159,047, which is a dramatic outlier compared to the median of 132. The minimum is 0, meaning some posts received no comments at all. The standard deviation is 3,569, which is approximately seven (7) times the mean, which demonstrates significant variability across the dataset. When compared to shares and reactions, comment counts tend to be lower overall, which is expected since commenting requires more effort from users than simply pressing one button to react / share a post. Like the other two variables, the outliers in comment count are meaningful rather than to be ignored; high comment posts likely represent controversial or highly engaging content that sparked active discussion. There are only 2 records that have missing comment count values, therefore they can be ignored due to their lack of significance on the dataset.

**Part 2: Data Visualization: [Tableau]**

**Choice #1: Pie Chart – Post Type:**

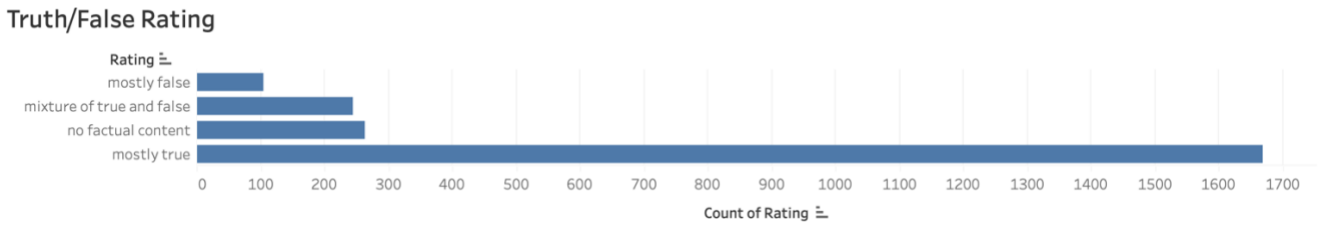



The pie chart displays the distribution of Facebook posts by post type across the dataset of 2,282 records. Link posts are dominating the dataset; they account for 1,780 posts (78.0% of the total). Video posts are the second most common at 291 posts (12.8%), followed by photo posts at 207 (9.1%), and text posts are the smallest category with only 4 posts (0.2%). This suggests that the majority of political news content on

Facebook during this period was shared by linked articles; this is consistent with how news organizations typically distribute their content. The relatively small number of video posts is notable because video content tends to generate significantly higher engagement; in this data set, photo posts averaged 16,038 shares and 25,102 reactions compared to just 1,222 shares and 2,404 reactions for link posts. This suggests that while link posts are the primary format for news distribution, visual content such as photos and videos may be more effective at driving audience engagement. T

**Choice #2: Bar Chart - Different Types of Ratings of Falsehood**




This Bar chart shows the count of Facebook posts across four different truth/falsehood ratings. The “mostly true” category dominates with approximately 1,669 posts (73.1%), making it by far the largest bar on the chart. The next two categories are much closer in size: ‘no factual content’ at 264 posts (11.6%) and ‘mixture of true and false at 245 posts (10.7%). The smallest category is ‘mostly false’ with only 104 posts (4.6%). The key finding is that most political news on Facebook during this time period were rated as substantially truthful. However, the data also shows that roughly 15.3% of posts contained some degree of falsehood when combining ‘mostly false’ and ‘mixture of true and false, which represents a meaningful share of political content on the platform. The ‘no factual content’ category (11.6%) is also significant, as these posts consist of opinion, commentary, or editorial content that cannot be fact-checked against verified claims. An interesting pattern emerges when considering engagement: posts rated as ‘no factual content’ generated the highest average share count (19,648 shares) and reaction count (19518 reactions), far exceeding the averages for ‘mostly true’ posts (1676 shares and 2,964 reactions). This suggests that opinion driven and emotionally charged content may resonate more strongly with Facebook for audiences than straightforward factual reporting, which has important implications for understanding misinformation dynamics on social media.

## Conclusion:

This analysis of 2,282 Facebook fact-check records from September 2016 revealed several key insights about political news content on the platform. Across all three-engagement metrics, share count, reaction count, and comment count is the data with extreme right-skewed distributions, where most posts received modest engagement while a small number of viral posts generated outsized activity. The pie chart of post types showed that link posts dominated at 78% of all content, yet photo and video posts tended to drive significantly higher engagement, suggesting that visual media is more effective at capturing audience attention. The bar chart of falsehood ratings revealed that while many posts (73.1%) were rated as "mostly true," approximately 15.3% contained some degree of falsehood, and the "no factual content" category, which is representing opinion and commentary and has generated the highest average shares and reactions of any rating type. Taken together, these findings suggest that emotionally driven and opinion-

based content tends to outperform straightforward factual reporting in terms of audience engagement on Facebook, which raises important questions about how social media platforms amplify certain types of political content over others.